

Lab: Ethics, Values, and the Normative Dimension

Learning Goal: Apply philosophical analysis to the ethical implications of Active Inference.

Part 1: The Is-Ought Analysis

Exercise: Evaluate three attempts to bridge the is-ought gap using Active Inference:

Attempt	Argument	Strength (1-5)	Weakness
Autopoietic values: "To be a living system <i>is</i> to value homeostasis — existence itself grounds values"	Bridges is-ought through identity: being = valuing		The transition from "maintains homeostasis" to "ought to maintain homeostasis" is unclear
Epistemic imperative: "A system that minimizes expected free energy <i>must</i> be curious and open-minded — intellectual virtue is built into the math"	EFE's epistemic term grounds a virtue (openness to evidence)		Curiosity can serve bad ends; openness $\neq$ goodness
Coherence norm: "Incoherent preferences generate unnecessary suffering (high free energy) — coherence is inherently preferable"	Internal consistency = well-being; inconsistency = suffering		Many coherent systems are morally monstrous (a perfectly coherent psychopath)

Write a 400-word analysis: Can any of these arguments successfully bridge the is-ought gap? If not, what does this tell us about the limits of Active Inference as an ethical framework?

Write your response here

Part 2: The C Vector Debate

Learning Goal: Analyze the philosophical status of preferences.

Exercise: Consider these scenarios:

Scenario A: An agent's C vector encodes strong preferences for helping others, learning, and social connection. The agent acts accordingly and experiences low free energy.

Scenario B: An agent's C vector encodes strong preferences for dominating others, hoarding resources, and avoiding uncertainty. The agent acts accordingly and experiences low free energy.

Both agents are equally good at minimizing free energy. Both are "rational" within their own models.

1. Is there any basis within Active Inference to say Agent A's preferences are *better* than Agent B's?
2. If we invoke external criteria (evolutionary fitness, social harmony, human flourishing), are these part of Active Inference or supplements to it?
3. How does this relate to the philosophical debate between moral realism and moral relativism?

Write your response here

Part 3: AI Ethics Case Study

Learning Goal: Apply Active Inference to a concrete AI ethics problem.

Exercise: You are designing an Active Inference-based AI assistant. Address these design decisions:

1. **Value specification:** How do you set the C vector? Who decides what the AI should prefer?
2. **Autonomy:** Should the AI's generative model be allowed to develop its own preferences through learning? What are the risks?
3. **Transparency:** How does Active Inference's interpretability help or hinder ethical oversight?
4. **Moral status:** If the AI develops self-modeling capabilities (Module 02), does it acquire moral claims? Should we care about its "free energy"?
5. **Shutdown problem:** An Active Inference agent that models its own continued existence may resist shutdown (shutting down = high surprise). How do you handle this?

Write a 400-word ethical analysis of your design choices.

Write your response here

Part 4: Political Philosophy Application

Learning Goal: Apply Active Inference to analyze a political phenomenon.

Exercise: Choose one of the following and analyze it through Active Inference's political philosophy:

Option A: Propaganda and precision control — How does state propaganda manipulate the precision of citizens' prediction errors to control beliefs?

Option B: Democratic deliberation — How does democratic debate function as collective Active Inference? When does it succeed (convergence toward truth) and when does it fail (polarization, echo chambers)?

Option C: Institutional trust — How is trust in institutions (government, science, media) a form of prior weighting? What happens when trust collapses?

Write a 400-word analysis of your chosen topic using Active Inference concepts.

Write your response here

Part 5: Your Ethical Position

Learning Goal: Synthesize and articulate your own philosophical position.

Exercise: Having studied eight modules of philosophical foundations, articulate your position on the following:

1. What can Active Inference contribute to ethics that other frameworks cannot?
2. What are the limits of Active Inference as an ethical framework?
3. How does Active Inference change (or not change) your understanding of values, responsibility, and the good life?

Write a 500-word philosophical position paper.

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Logical analysis	Is-ought problem
2	Thought experiment	Preference evaluation
3	Applied ethics	AI design decisions
4	Political analysis	Power, deliberation, trust
5	Synthesis	Personal philosophical position